

# Minimum-Variance Portfolios and the Capital Allocation Line

## LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Formulate portfolio risk.** Express expected growth rate and variance as functions of weights, asset means, and the covariance matrix.
- **Construct the frontier.** Solve global and target-return minimum-variance problems and distinguish the feasible set from its efficient boundary.
- **Add the risk-free asset.** Derive the capital allocation line and tangency portfolio and state the assumptions behind calling it the market portfolio.

## 1 Diversification Is a Covariance Calculation

Harry Markowitz's central insight was that an asset should not be judged in isolation. A volatile asset can reduce portfolio variance when its growth rates offset other holdings, while individually stable assets can provide little diversification when they move together.

Let  $\mathbf{g} \in \mathbb{R}^m$  be a random vector of risky-asset growth rates, with mean  $\bar{\mathbf{g}} := \mathbb{E}[\mathbf{g}]$  and positive-definite covariance matrix  $\Sigma := \text{Cov}(\mathbf{g})$ . Let  $\mathbf{w} \in \mathbb{R}^m$  be risky-asset dollar weights satisfying  $\mathbf{1}^\top \mathbf{w} = 1$ . The portfolio growth rate  $g_p := \mathbf{w}^\top \mathbf{g}$  has

$$\mathbb{E}[g_p] = \mathbf{w}^\top \bar{\mathbf{g}}, \quad \sigma_{g,p}^2 = \mathbf{w}^\top \Sigma \mathbf{w}. \quad (1)$$

The cross terms  $2w_i w_j \Sigma_{ij}$  create diversification effects. Growth-rate covariance has units of inverse time squared; all inputs must use a common time convention. The associated log return over  $\Delta t$  is  $r_p = (\Delta t)g_p$ .

For a fixed observation interval, let  $\Sigma_r := \text{Cov}(\mathbf{r})$ ,  $\Sigma_g := \text{Cov}(\mathbf{g})$ , and let  $C$  denote the diffusion covariance rate used by GBM. Because  $\mathbf{r} = (\Delta t)\mathbf{g}$ ,

$$\Sigma_r = (\Delta t)^2 \Sigma_g, \quad C = \frac{\Sigma_r}{\Delta t} = \Delta t \Sigma_g. \quad (2)$$

This lecture uses  $\Sigma = \Sigma_g$  throughout the allocation problems. Replacing it by  $\Sigma_r$  merely scales an unconstrained variance objective by  $(\Delta t)^2$ , but means and targets must also satisfy  $\bar{\mathbf{r}} = (\Delta t)\bar{\mathbf{g}}$  and  $r_* = (\Delta t)g_*$ . The covariance rate  $C$  is reserved for GBM simulation and annualized-volatility reporting. For daily observations with  $\Delta t = 1/252$ , the diffusion model implies  $\Sigma_g = 252C$ . Thus growth-rate variances can look numerically “huge” compared with conventional annual covariance-rate entries; that factor is the required unit conversion, not an estimation failure. Portfolio weights are unchanged by this common positive rescaling when every other constraint is expressed consistently, but reported risk and Sharpe-ratio horizons are not interchangeable.

## 2 The Global Minimum-Variance Portfolio

The global minimum-variance (GMV) portfolio selects the fully invested risky allocation with the smallest modeled variance. When short positions are allowed and  $\Sigma$  is positive definite, the equality-constrained problem has a closed form (Thm. 2.1).

### Theorem 2.1. Unconstrained global minimum-variance portfolio

Let  $\Sigma \in \mathbb{R}^{m \times m}$  be symmetric positive definite and let  $\mathbf{1} \in \mathbb{R}^m$  be the vector of ones. The unique solution of

$$\min_{\mathbf{w} \in \mathbb{R}^m} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1 \quad (3)$$

is

$$\mathbf{w}_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad \sigma_{\text{GMV}}^2 = \frac{1}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}. \quad (4)$$

The solution can contain negative weights because no long-only constraint is imposed.

**Derivation.** Introduce the Lagrange multiplier  $\lambda \in \mathbb{R}$  for the budget constraint:

$$\mathcal{L}(\mathbf{w}, \lambda) = \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} - \lambda (\mathbf{1}^\top \mathbf{w} - 1).$$

Stationarity gives  $\Sigma \mathbf{w} - \lambda \mathbf{1} = \mathbf{0}$ , so  $\mathbf{w} = \lambda \Sigma^{-1} \mathbf{1}$ . Enforcing  $\mathbf{1}^\top \mathbf{w} = 1$  gives  $\lambda = (\mathbf{1}^\top \Sigma^{-1} \mathbf{1})^{-1}$  and yields Eq. 4.  $\square$

Imposing  $w_i \geq 0$  converts the problem to a convex quadratic program. Its optimum may lie on a boundary with some weights exactly zero, and the unconstrained formula is no longer generally valid. Other realistic constraints—position limits, turnover, lots, taxes, and transaction costs—change the feasible set further.

## 3 The Minimum-Variance Frontier

To trace growth–risk tradeoffs, add a target expected growth rate  $g_*$ :

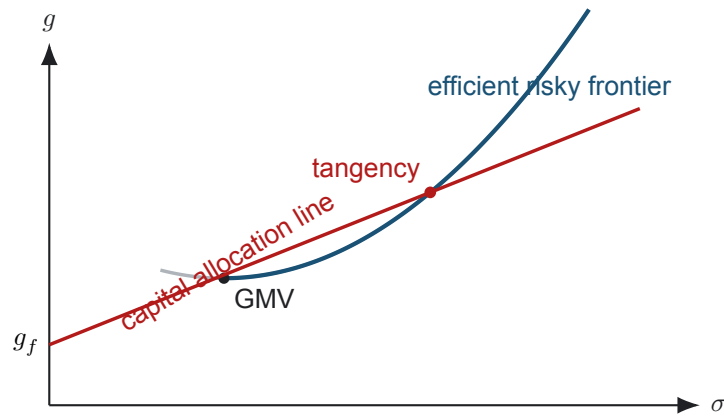
$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \bar{\mathbf{g}}^\top \mathbf{w} = g_*. \quad (5)$$

Varying  $g_*$  produces the minimum-variance frontier. The branch above the GMV expected growth rate is the *efficient frontier*: no other feasible risky portfolio has both at least as much expected growth and no more variance, with one strict improvement.

### COMPANION NOTEBOOK

#### L5b: Data-Driven Minimum-Variance Portfolios →

Estimate means and covariance, solve long-only target-return quadratic programs, and visualize the empirical efficient frontier.



**Figure 1:** The risky-asset minimum-variance frontier and a capital allocation line from a risk-free growth rate  $g_f$ . The line with the largest attainable slope is tangent to the risky frontier.

## 4 Adding a Risk-Free Asset

Let a locally risk-free asset earn deterministic growth rate  $g_f$ . Combine it with any risky portfolio having expected growth rate  $g_p$  and standard deviation  $\sigma_p > 0$ . If  $y \in \mathbb{R}$  is the dollar fraction invested in that risky portfolio, then

$$g_c = g_f + y(g_p - g_f), \quad \sigma_c = |y|\sigma_p. \quad (6)$$

For  $y \geq 0$ , eliminating  $y$  gives the capital allocation line (CAL):

$$g_c = g_f + \frac{g_p - g_f}{\sigma_p} \sigma_c. \quad (7)$$

The slope is the risky portfolio's Sharpe ratio under the stated horizon and risk-free convention. Values  $0 < y < 1$  lend some wealth at  $g_f$ ;  $y = 1$  holds only the risky portfolio; and  $y > 1$  represents leveraged risky exposure financed by borrowing. A single identical lending and borrowing rate is an idealization.

Among all risky portfolios, the tangency portfolio maximizes the CAL slope. Without long-only constraints, its weights have the closed form in [Prop. 4.1](#).

### Proposition 4.1. Unconstrained tangency portfolio

Let  $\Sigma$  be symmetric positive definite, let risky expected growth rates be  $\bar{\mathbf{g}} \in \mathbb{R}^m$ , and let the risk-free growth rate be  $g_f \in \mathbb{R}$ . Assume  $\bar{\mathbf{g}} - g_f \mathbf{1} \neq \mathbf{0}$ . The fully invested risky portfolio proportional to  $\Sigma^{-1}(\bar{\mathbf{g}} - g_f \mathbf{1})$  has weights

$$\mathbf{w}_T = \frac{\Sigma^{-1}(\bar{\mathbf{g}} - g_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\bar{\mathbf{g}} - g_f \mathbf{1})}. \quad (8)$$

It maximizes the positive Sharpe ratio among unconstrained fully invested risky portfolios under these assumptions.

Every mean-variance investor facing the same inputs can then combine the same tangency fund with the risk-free asset, choosing  $y$  according to risk tolerance. This is the two-fund separation result. Long-only, leverage, and borrowing constraints can kink the attainable boundary and change the tangency solution.

## 5 When Is the Tangency Portfolio the Market Portfolio?

In the Capital Asset Pricing Model, all investors share the same beliefs, optimize mean and variance over the same horizon, can borrow and lend at the same risk-free rate, and trade all assets without taxes or frictions. Market clearing then identifies the aggregate value-weighted risky portfolio with the common tangency portfolio. This theoretical object is called the *market portfolio*.

An optimizer applied to a selected ticker list and historical estimates does not thereby recover the market portfolio. It produces a sample-dependent tangency portfolio for that universe and model. A broad stock index also omits many risky assets. The CAPM identification is an equilibrium conclusion under strong assumptions, not a label to attach to every maximum-Sharpe solution.

## 6 Estimation Risk Can Dominate Optimization

The inverse covariance in [Eq. 4](#) and [Eq. 8](#) amplifies unstable directions. Tangency weights are especially sensitive to expected-return estimates, which are noisy. Backtests must fit inputs only on information available before each decision, include turnover and costs, and compare with simple baselines such as equal weighting. Covariance shrinkage, weight constraints, and robust objectives can improve stability but change the optimization problem and must be reported.

The MGBM notebook supplies a separate scenario engine. It can test how a fixed allocation behaves under the assumed joint dynamics, but simulated performance is not independent validation when the same data estimated both the optimizer and simulator.

### COMPANION NOTEBOOK

#### **L5b: MGBM Portfolio Simulation** →

Simulate correlated asset prices and compare a chosen allocation's wealth distribution with actual and benchmark performance.

**KEY TAKEAWAYS**

In this lecture, we derived minimum-variance weights, traced the efficient frontier, and added a risk-free asset to obtain the CAL and tangency portfolio.

- **Covariance determines diversification.** Portfolio variance is  $w^T \Sigma w$ ; cross-asset terms determine whether combining assets reduces modeled risk.
- **The CAL selects a risky fund.** The tangency portfolio maximizes reward per unit of modeled risk, while the risk-free allocation controls an investor's overall exposure.
- **Market portfolio is an equilibrium claim.** Calling tangency weights the market portfolio requires the CAPM universe, common-belief, trading, and market-clearing assumptions.

Next, replace the full covariance description with a single-index factor model and separate systematic from idiosyncratic risk.

**ADDITIONAL READING**

- Harry Markowitz, “Portfolio Selection,” *Journal of Finance* 7(1), 77–91 (1952).
- William F. Sharpe, “Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk,” *Journal of Finance* 19(3), 425–442 (1964).
- Richard O. Michaud, *Efficient Asset Management*, 2nd ed. (2008), for estimation error and instability in mean–variance optimization.

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